



US012402284B2

(12) **United States Patent**
Har-Shai et al.

(10) **Patent No.:** **US 12,402,284 B2**

(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **COOLING MECHANISM FOR
MULTI-COMPARTMENT POWER DEVICE**

USPC 361/714
See application file for complete search history.

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(56)

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 1 day.

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(21) Appl. No.: **18/528,186**

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(22) Filed: **Dec. 4, 2023**

KR 20180080927 A 7/2018

(65) **Prior Publication Data**

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Dec. 6, 2021—European Search Report—EP App. No. 21182041.0.

Related U.S. Application Data

Primary Examiner — Anatoly Vortman

(63) Continuation of application No. 17/363,925, filed on
Jun. 30, 2021, now Pat. No. 11,871,543.

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(60) Provisional application No. 63/045,940, filed on Jun.
30, 2020.

(57)

ABSTRACT

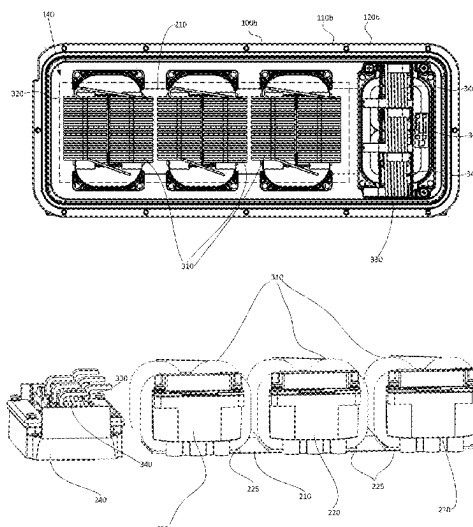
(51) **Int. Cl.**
H05K 7/20 (2006.01)
H01F 27/08 (2006.01)

Inverters for direct current (DC) to alternating current (AC)
conversion may comprise switching elements which pro-
duce harmonic overtones. Inductive elements may be added
into the inverters, such as to attenuate effects of harmonic
overtones. Methods and systems for a casing suitable for
reduced potting over its inductive elements is described.
One or more heat dispersing elements may be disposed in
the casing. Related systems and methods are also described.

(52) **U.S. Cl.**
CPC **H05K 7/20418** (2013.01); **H01F 27/08**
(2013.01); **H05K 7/209** (2013.01)

(58) **Field of Classification Search**
CPC H05K 7/20418; H05K 7/209; H01F 27/08

20 Claims, 10 Drawing Sheets



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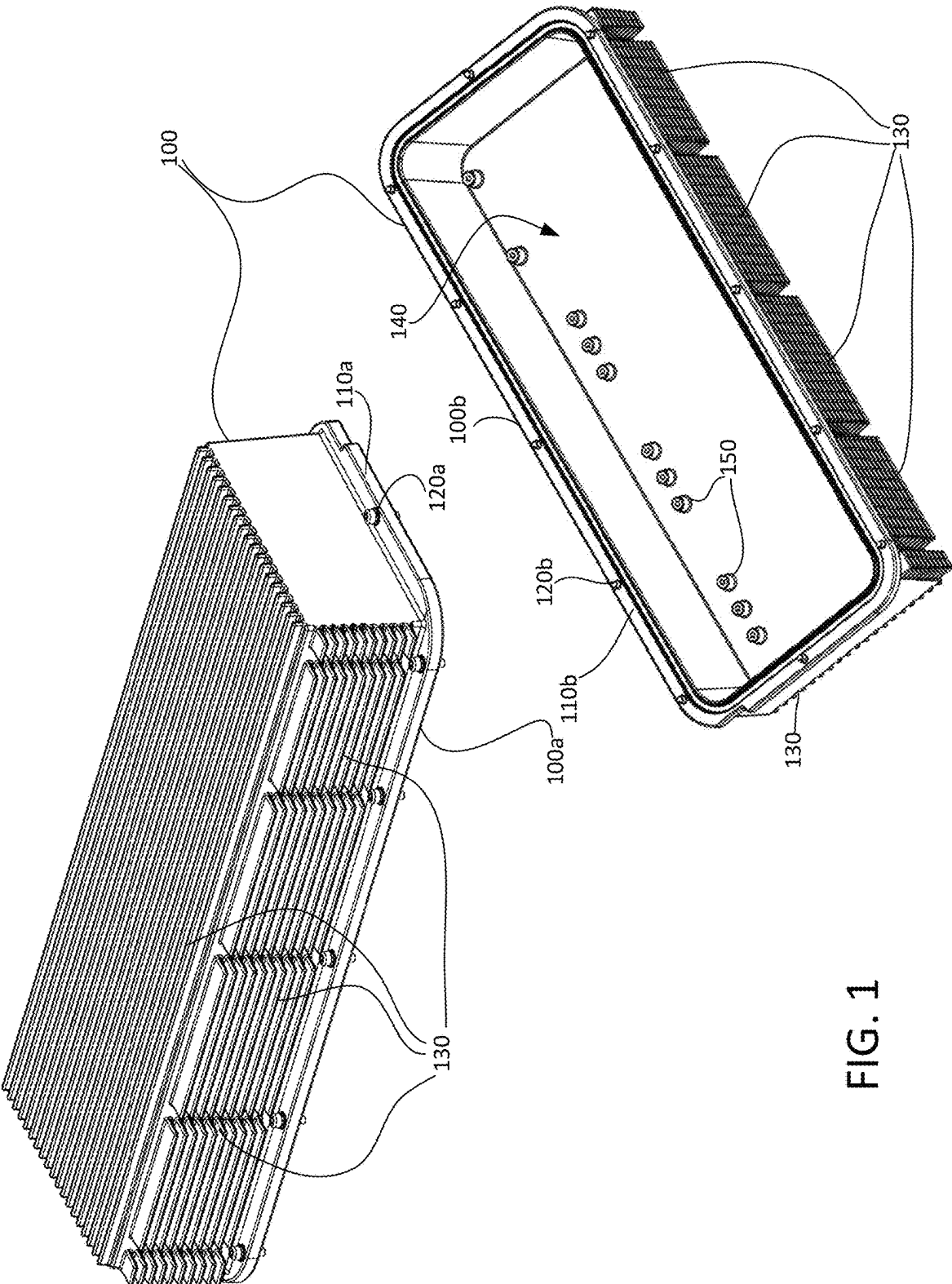


FIG. 1

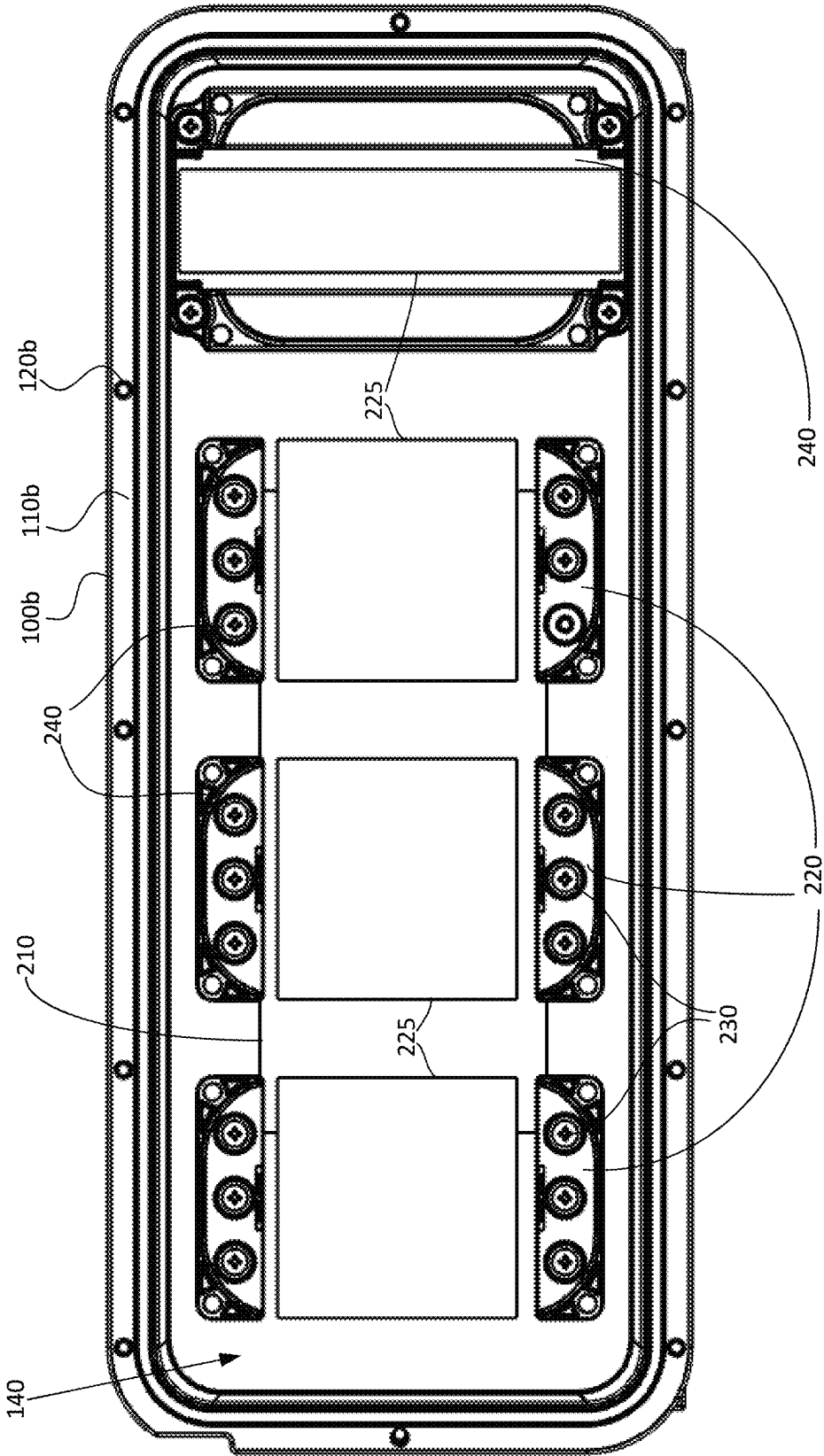


FIG. 2

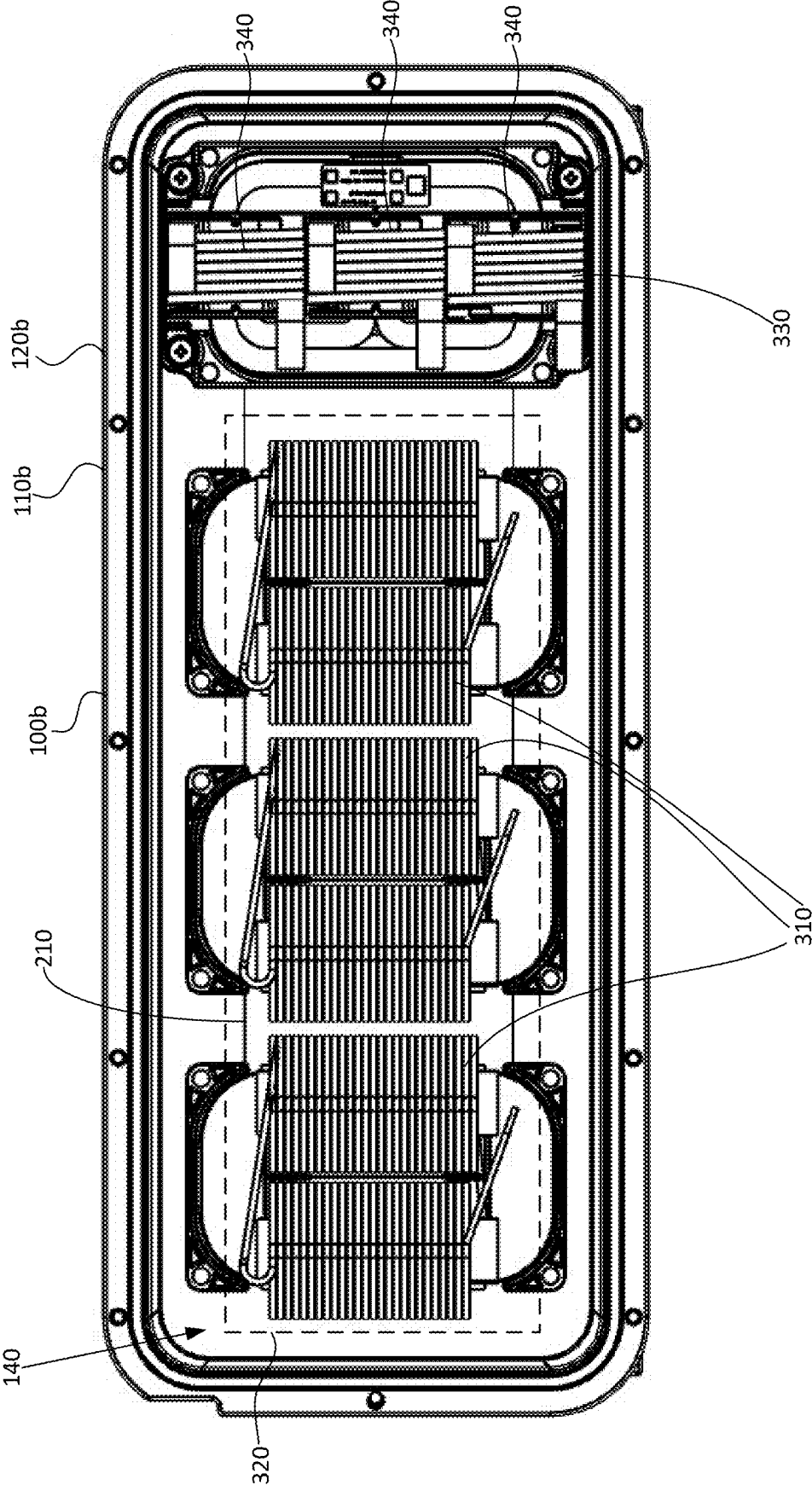


FIG. 3

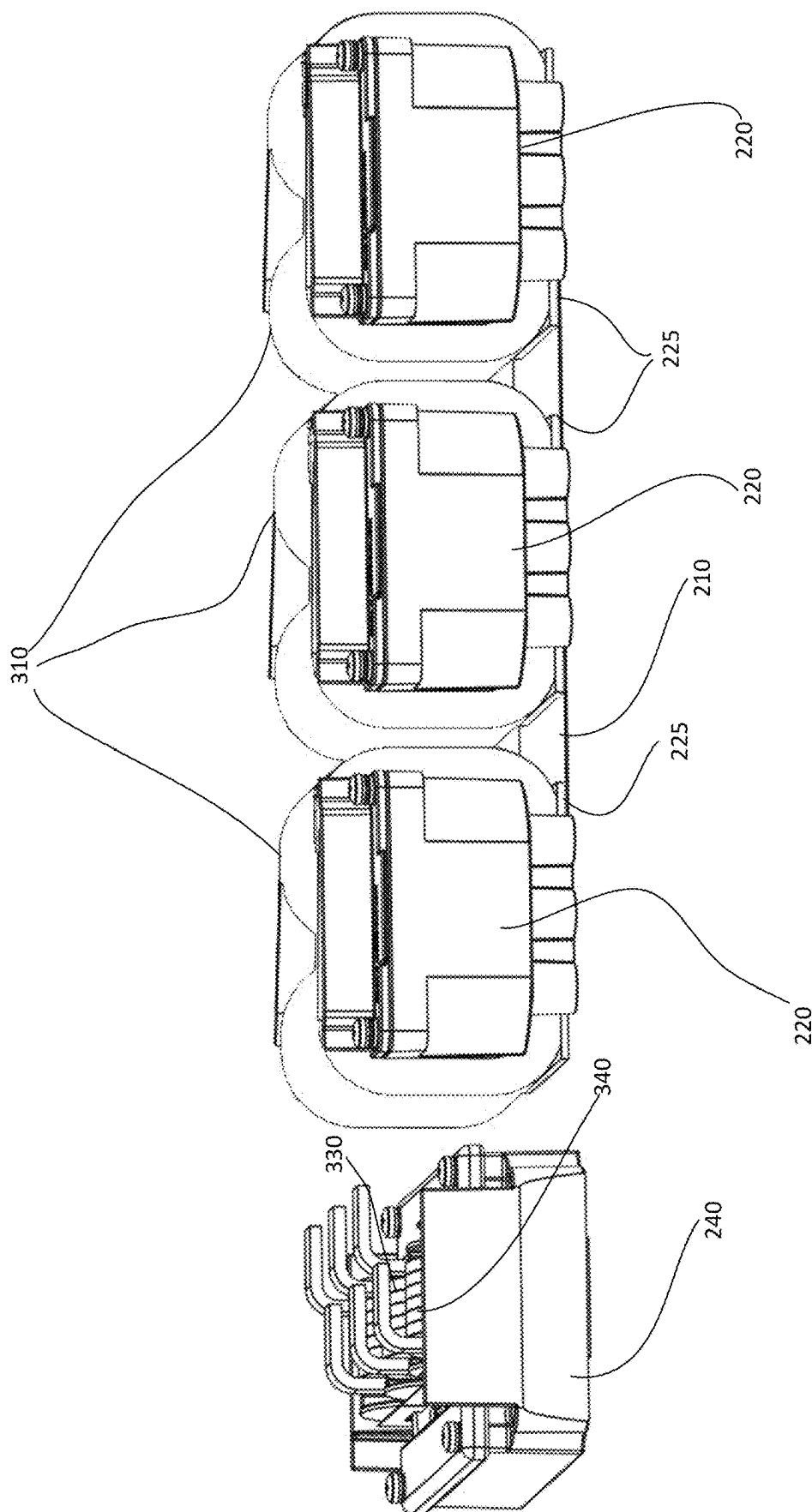
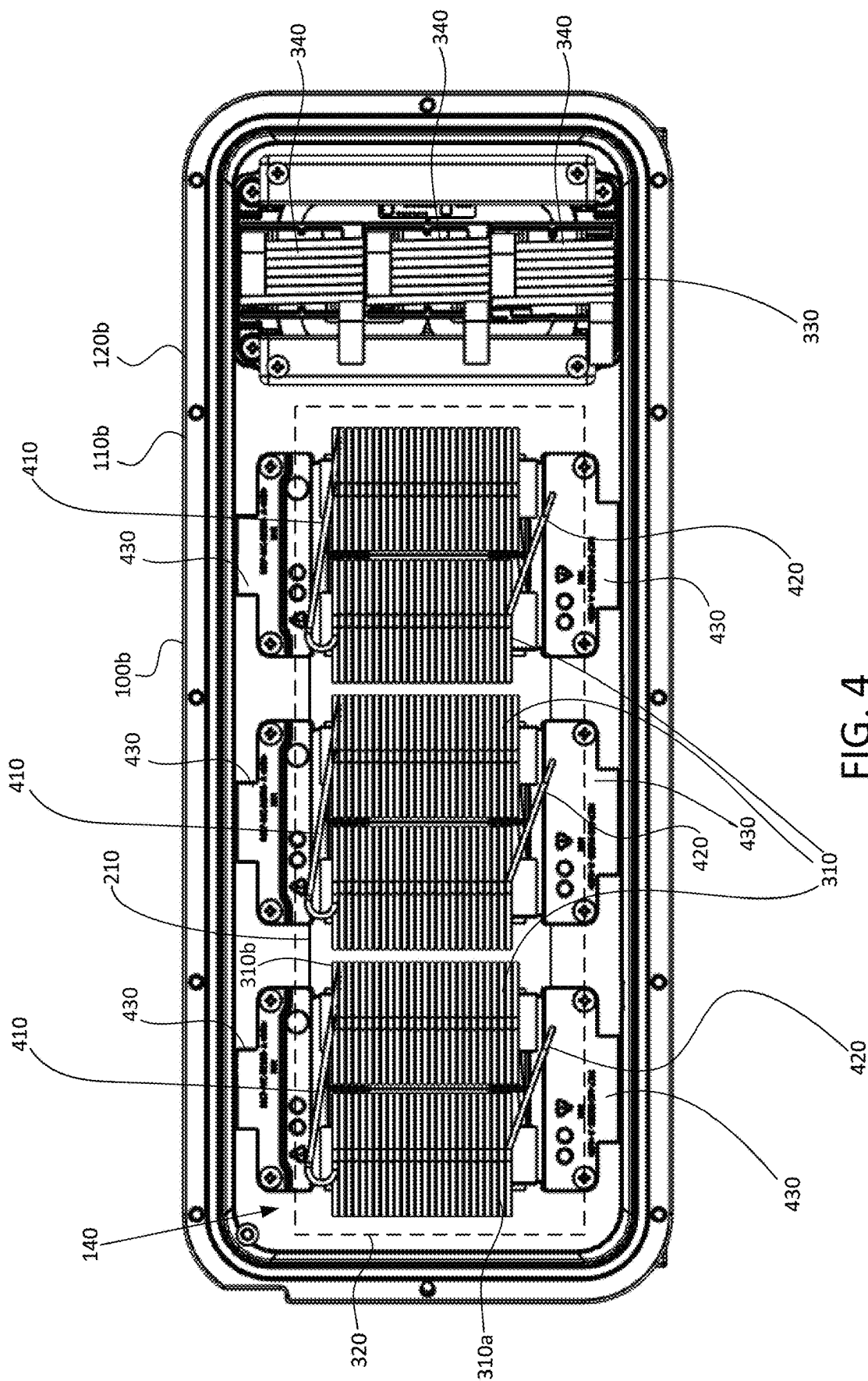


FIG. 3A



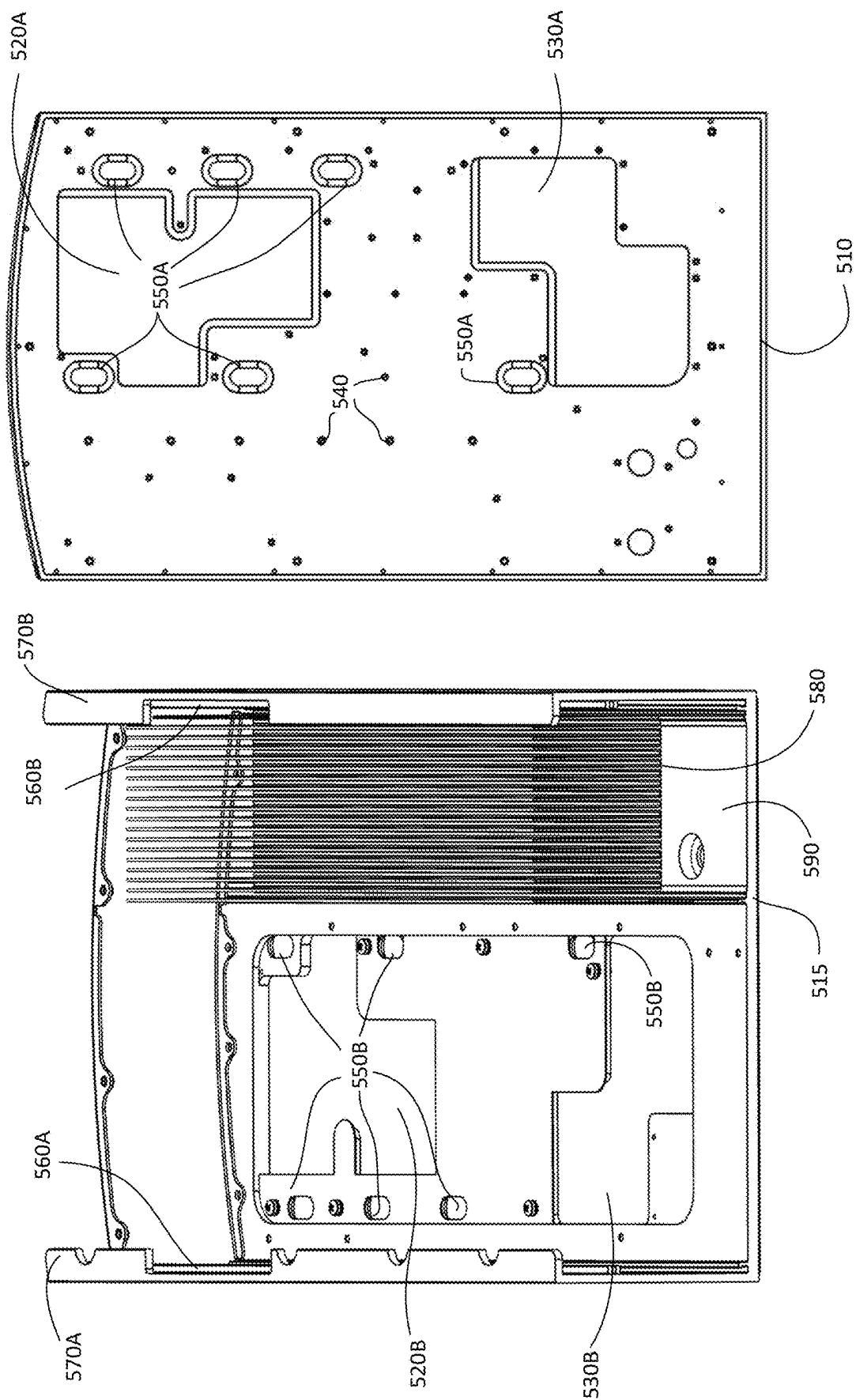


FIG. 5

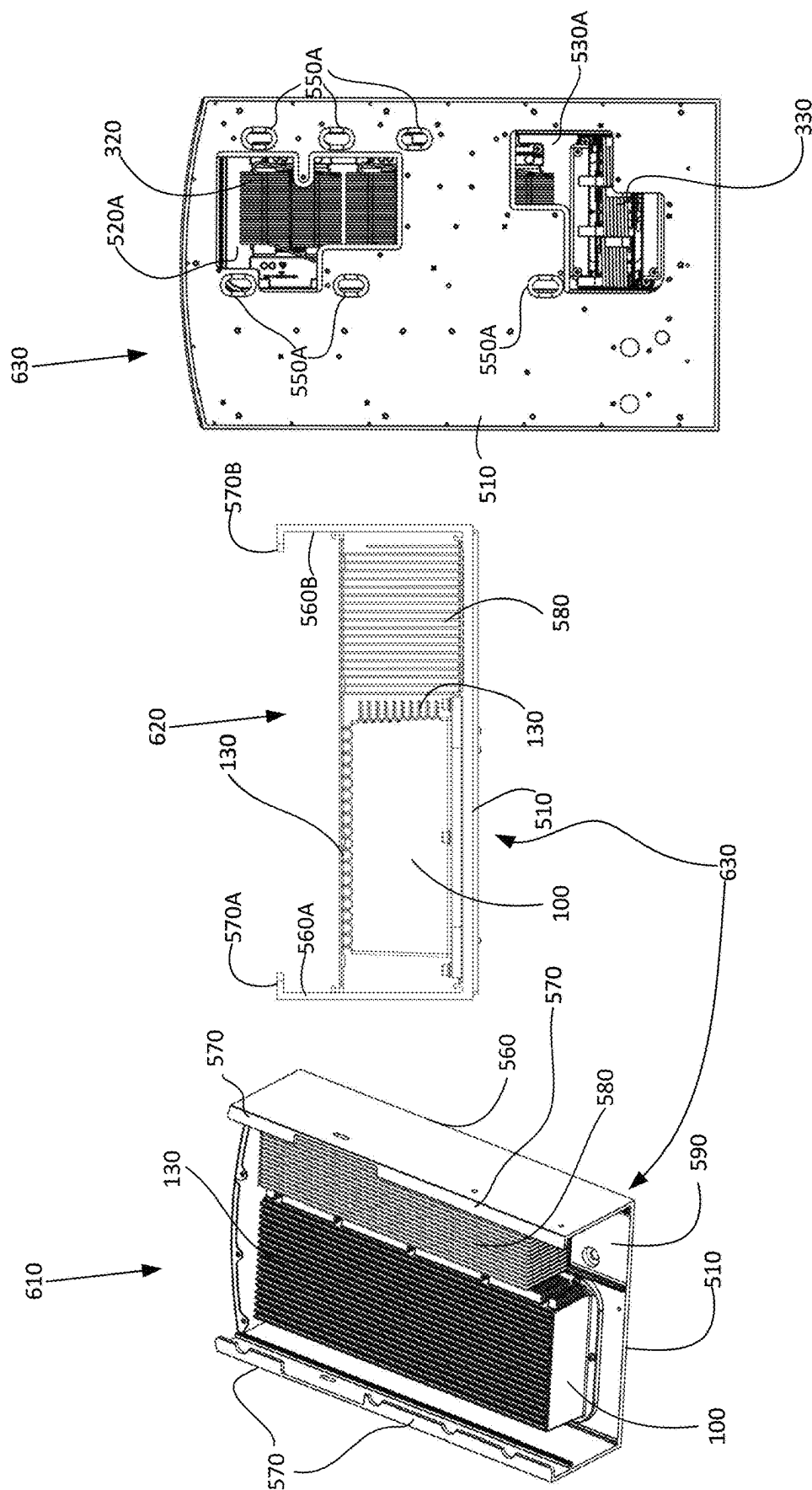
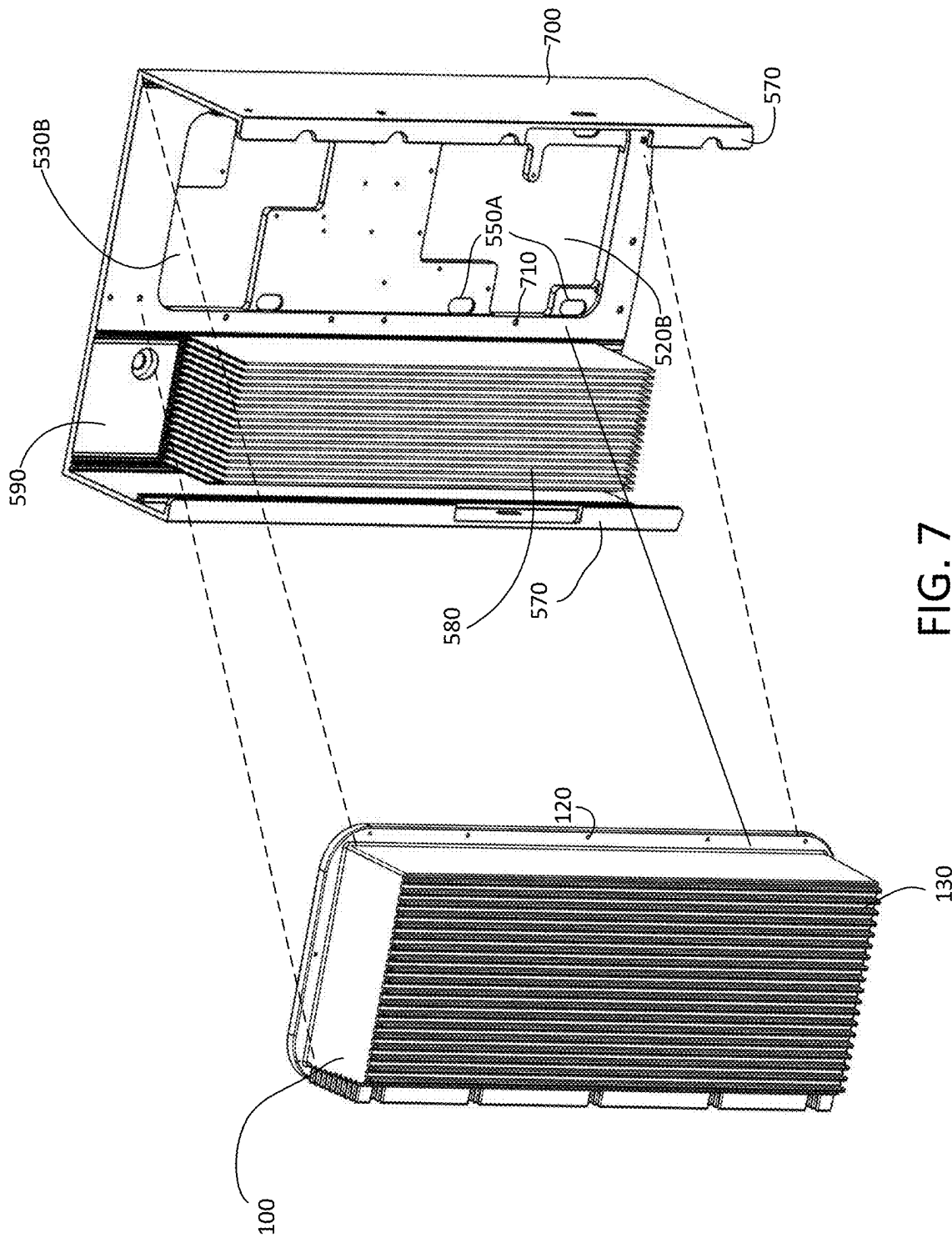


FIG. 6



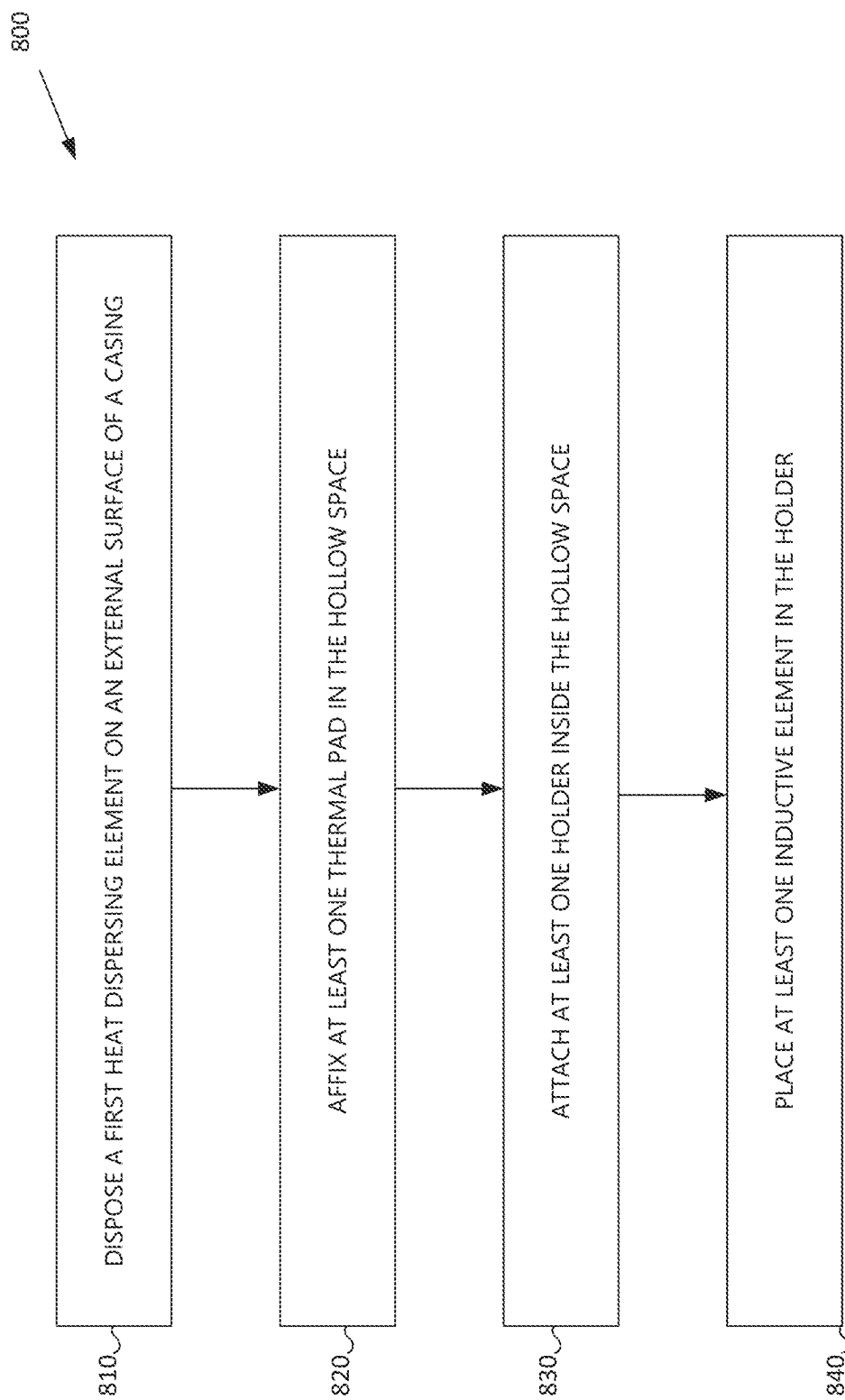


FIG. 8

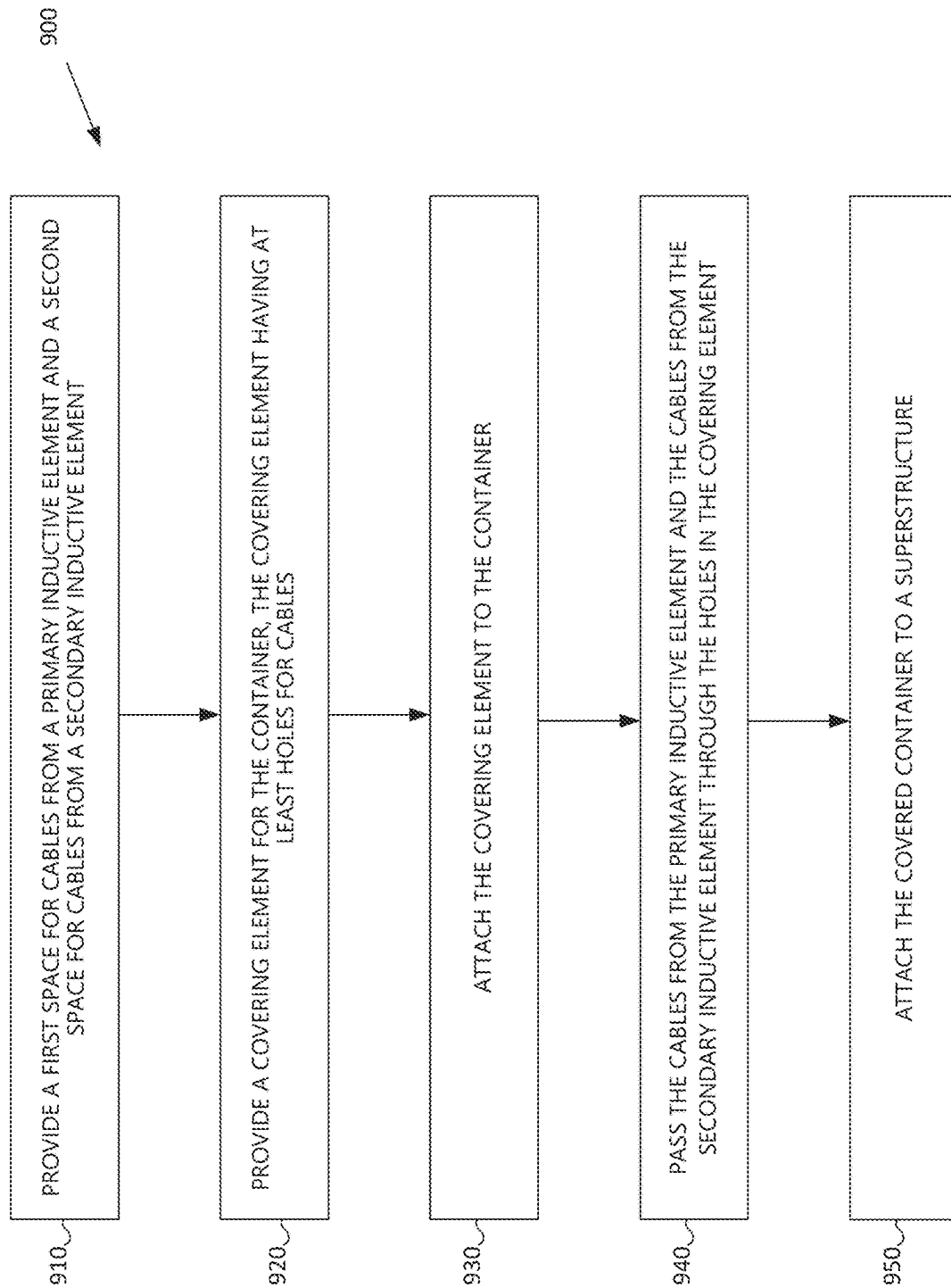


FIG. 9

1

COOLING MECHANISM FOR MULTI-COMPARTMENT POWER DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 17/363,952, filed Jun. 30, 2021, which claims priority to U.S. Provisional Patent Application No. 63/045,940, filed Jun. 30, 2020. The contents of the above identified application are incorporated herein by reference in its entirety.

BACKGROUND

An electronic assembly may be filled, partially or completely, with a solid or gelatinous compound. Such a filling, or 'potting', may provide increased heat conductance, resistance to shock and vibration, exclusion of gaseous phenomena such as corona discharge, or protection from environmental elements such as water or potentially corrosive chemicals. Thermosetting plastics, silicone rubber gels, or epoxy resins are often used as potting materials.

Potting often entails placing an electronic assembly inside a mold (the 'pot') which is then filled with a liquid potting substance which then hardens as it dries, thereby providing a measure of long term protection to the assembly. The pot may remain with the assembly or not, depending on particular product requirements.

A disadvantage of the potting process is the cost and weight of the potting compound. When manufacturing large numbers of units which require potting, each individual unit may require large quantities of potting compound. Multiplying by a large number of mass produced units may quickly turn the potting compound into a major expense in the manufacturing process, as well as a major factor in the final weight of the product including the potted unit.

SUMMARY

The following summary presents a simplified summary of certain features. The summary is not an extensive overview and is not intended to identify key or critical elements.

A casing having a first heat dispersing element disposed superficially on an external surface of the casing is described. The casing may have a hollow space formed by an internal surface of the casing. At least one thermal pad may be disposed in the hollow space. At least one holder may be disposed in the hollow space. At least one inductive element may be placed in the holder, the at least one inductive element disposed to be held by the holder such that the at least one inductive element is held over the at least one thermal pad.

The casing may be configured to be disposed in and/or mechanically connected to a superstructure. The superstructure may comprise a second heat dispersing element disposed superficially on an external surface of the superstructure.

The first heat dispersing element may comprise fins. The second heat dispersing element also may comprise fins. The at least one holder may be made of an electrically isolating material, for example, plastic. The inductive element may comprise, for example, an inductor coil, a choke, a reactor, a transformer.

The superstructure may comprise a casing of a direct current (DC)-to-alternating current (AC) inverter mechanism. The superstructure may comprise a casing of an AC to

2

DC power converter. The superstructure may comprise a casing of a DC-to-DC converter.

An apparatus which does not have potting over its inductive elements is described, as well as methods for constructing such the apparatus. An apparatus which does not have potting over heat generating elements is described, as well as methods for constructing such the apparatus. An apparatus which does not have potting over electrical conducting elements is described, as well as methods for constructing such the apparatus.

These and other features and advantages are described in greater detail below.

BRIEF DESCRIPTION OF THE DRAWINGS

Some features are shown by way of example, and not by limitation, in the accompanying drawings. In the drawings, like numerals reference similar elements.

FIG. 1 shows a first view and a second view of a casing.

FIG. 2 shows a view of part of a casing.

FIG. 3 shows a view of a casing with inductive elements.

FIG. 3A shows a side view with inductive elements.

FIG. 4 shows a detail of a casing with supportive extrusions.

FIG. 5 shows a first covering element and second covering element.

FIG. 6 shows different perspectives of elements of the superstructure.

FIG. 7 shows a structural relationship between a casing and a frame which may hold the casing and the superstructure together.

FIG. 8 is a flow diagram showing an example method for assembling a mechanism included in the casing.

FIG. 9 is a flow diagram showing an example method for attaching the container of a casing to the superstructure.

DETAILED DESCRIPTION

The accompanying drawings, which form a part hereof, show examples of the disclosure. It is to be understood that the examples shown in the drawings and/or discussed herein are non-exclusive and that there are other examples of how the disclosure may be practiced.

Reference is now made to FIG. 1, which shows a first view **100a** and a second view **100b** of a casing **100**. The casing **100** may be for use, for example, inside and/or attached to a power converter, for example, a direct current (DC) to alternating current (AC) power converter, an AC to DC power converter, or a DC-to-DC power converter. The first view **100A** may generally be considered an "external" view of the casing **100**, and the second view **100b** may generally be considered an "internal" view of the casing **100**. The first view **100a** may be referred to below, in a non-limiting fashion, as "the external view" of the casing **100a**. The second view **100b** may be referred to below, in a non-limiting fashion, as "the internal view" of the casing **100a**. The same or a similar element may be depicted in one or the other of the external view **100a** and the internal view **100b**. In such cases, the element's reference number will often be provided with the suffix "a" in the case of the external view **100a**, and the suffix "b" in the case of the internal view **100b**. In cases where the external view **100a** and the internal view **100b** of a particular element does not substantially differ, then no distinction will be noted with the use of the suffix.

The casing **100** may comprise a lip **110a** and/or a lip **110b**. The lip **110a** or **110b** may comprise respective holes **120a** or

3

120b through which fastening elements, for example, bolts, rivets, screws, etc. may be passed, in order to fasten the casing 100 to a superstructure, as described below with reference to FIGS. 5-7.

The casing 100 may comprise a first heat dispersing element 130 disposed superficially on an external surface of the casing 100. The first heat dispersing element 130 may comprise fins. The fins may increase surface area of the casing 100, which may allow for greater exposure of the casing 100 to surrounding air (or other surrounding media). The air may serve as a cooling medium with which heat from the first heat dispersing element 130 may be exchanged.

The casing 100 may comprise an internal portion 140. The internal portion 140 may provide a hollow space, which may be formed by an internal surface of the casing 100. Various elements and/or apparatus to be described below may be placed or fitted in the internal portion 140 of the casing 100. The internal portion 140 may comprise spacers 150, into which fastening elements may be affixed for holding various apparatus in place in the internal portion 140, as may be described below.

Reference is now made to FIG. 2, which shows a detail of the second view 100b of the casing 100. The detail of the second view 100b of the casing 100 is showing insulating padding 210 and holders 220. The insulating padding 210, which may comprise an electrically and/or thermally insulating barrier (for example, Nomex® or Kapton®), may be disposed on an internal surface of the casing 100, or a portion of the internal surface of the casing 100. The insulating padding 210 may be disposed next to or near heat generating elements which are placed in the casing 100, as may be described below. A thermal pad 225 may be disposed at the base of one or more holders 220. The thermal pads may be made of a material including paraffin wax or silicone, and may generally conduct heat into a heat dispersing element (e.g., the first heat dispersing element 130, which may be disposed on the opposing side of the casing 100, as may be described above regarding FIG. 1).

It is appreciated that, due to the heat which may be generated inside the internal portion 140 during operation, it may be desirable to shed heat while also providing electrical insulation between the casing 100 and the elements disposed in the internal portion 140. This may be beneficial in the event that the internal portion 140 contains elements which conduct electricity. Thus, the actual insulating padding 210 chosen may be selected in order to provide a high degree of electrical insulation while providing a low degree of thermal insulation.

The casing 100 may comprise one or more holders 220, which may hold components of a DC-AC inverter which are electrically conductive and heat generating. Although FIG. 2 is described with reference to a DC-AC inverter, it is appreciated that the casing 100 may hold components of a DC-DC converter, an AC-DC converter, or other appropriate electronic and heat generating device. The holders 220 may be constructed of an appropriate plastic material, such as an insulating plastic, which may provide electrical and/or heat insulation. In some instances, fasteners used to couple one or more components, such as screws 230 (note that in order to reduce visual noise in the figure, not all screws 230 may be labeled as such) may be electrically conductive. Fasteners may be any suitable fastener, such as a screw, bolt, thermosatically molded coupling, glue, etc. Holders 220 may be designed to insulate conducted electricity of the screws 230. For example, holders 220 may be constructed of an insulating plastic or other appropriate insulating material. A

4

second holder 240 may optionally be provided in the internal portion 140 of the casing 100, for holding a second type of component. The second type of component may be a heat generating component, an electrically conductive component, both, or neither. At least one non-limiting example of a possible second component will be provided below.

A thermal pad 225 may be placed in each of the holders 220 or holders 240. The thermal pads 225 may aid the conduction of heat away from a heat generating component placed in the holder, as may be described below.

Throughout this disclosure and the accompanying figures, various elements may be depicted in a particular quantity. For example, three holders 220 are depicted in the casing 100, twelve holes 120b are depicted on the lip 110b, one differential choke block comprising three chokes (as may be depicted in FIG. 3, element 320) is depicted, and so forth. Such depictions and quantities are by way of example only, and not meant to be limiting. Other appropriate quantities of the various elements depicted may be utilized as a matter of course in actual implementations.

Reference is now made to FIG. 3, which may show a view consistent with FIG. 2 with one or more additional aspects. FIG. 3 shows the second view 100b, with inductive elements 310 in the casing 100. Internal portion 140 may form a first chamber configured to hold inductive elements 310. Inductive elements 310 may comprise any appropriate inductive element, such as an inductor, a reactor, a transformer etc. As may be described above, inductive elements 310 may be heat generating and/or electrically conductive inductive elements 310. According to features of the disclosure herein, instead of or in addition to inductive elements, other types of heat generating and/or electrically conductive elements may be disposed in internal portion 140, for example, conducting wires, transistors, resistors, diodes, or other electronic components.

A first group of inductive elements 320 (which may comprise at least one of the inductive elements 310) may be disposed in the casing 100, and may be held by the holders 220. A second group of inductive elements 330 may be disposed in the casing 100, and may be held by the holders 220. The second group of inductive elements 330 may comprise one or more secondary inductive elements 340. It is appreciated that any other appropriate heat generating and/or electrically conductive may be placed in the holders 220, and that the first group of inductive elements 320 and the second group of inductive elements 330 are only one example. In some instances, the first group of inductive elements 320 may include a main choke, and the second group of inductive elements 330 may include a secondary choke, for example, a filter choke (e.g., a common-mode choke and/or a differential choke). In some instances, the first group of inductive elements 320 and the second group of inductive elements 330 may be placed in different holders, such as holder 220. Appropriate concerns of design, space, cost, materials, etc. may affect particular implementations.

It is appreciated that although the above description and accompanying figures are speaking of chokes, choke blocks, inductive elements, etc., any appropriate heat-generating, and/or electrically conductive element may be disposed in the herein described, similar, or equivalent casing.

The various inductive elements 320 or 340 described herein may be held in the holders 220 so as to be disposed in proximity with the insulating padding 210. The insulating padding 210 may be disposed so as to direct heat generated by the at least one of the inductive element 310 and the at least one secondary inductive element 340 to the first heat

5

dispersing element **130**. The insulating padding **210** may insulate from electricity, which may be present in the various inductive elements **320** or **340**. The screws **230** may be manufactured out of a material which conducts electricity, but may be insulated by surrounding plastics from one or more conductive elements which may surround them.

Cabling (not depicted) from the second group of inductive elements **330** may be disposed on one side of the inductive elements **310**, and cabling (not depicted) from the first group of inductive elements **320** may be disposed on the other side of the inductive elements **310**. This may have the benefit of maintaining a physical separation between the cables from the first group of inductive elements **320** and the cables from the second group of inductive elements **330**.

FIG. **3A** shows a side view of inductive elements **310**, such as inductive elements **310** may be depicted in FIG. **3**. The insulating padding **210** is seen at the bottom of the figure. A plurality of the thermal pads **225** may be disposed above the insulating padding **210**. The holders **220** may be disposed at the sides of the insulating padding **210** and the thermal pads **225**. The inductive elements **310** may be disposed in the holders **220**, such that each inductive element **310** may be disposed above one of the thermal pads **225**. The second group of inductive elements **330** may be disposed in the second holder **240**. Insulating padding **210** and thermal pads **225** may also be disposed beneath the second group of inductive elements **330**.

Reference is now made to FIG. **4**, which may show a view consistent with FIG. **3** with one or more additional aspects, such as supportive extrusions **430**. The inductive elements **310** may comprise multiple portions of windings (e.g., first winding portion **310a** and second winding portion **310b**). Cabling **410** may connect, for example, between first winding portion **310a** and second winding portion **310b**, and may be supported by extrusions **430**. Extrusions **430** may comprise insulating plastic. Similarly, cabling leading to or from a terminal **420** into and/or out of the inductive element **310** may also be supported by extrusions **430**.

It is appreciated that the above discussion is showing one possible configuration for the elements depicted as being in the casing **100**. The one possible configuration depicted in FIG. **4** is not meant to be limiting, and is by way of example only. For example, as discussed above, there may be different numbers of inductive elements **310** or **340** in the casing **100**. Additionally, the first group of inductive elements **320** and the second group of inductive elements **330** may be disposed in different location than shown in the figures. The second group of inductive elements **330** may optionally not be present. The holders **220** or **240** may be in different locations, both with respect to the casing **100** and to each other. The insulating padding **210** may be applied to walls or other internal surfaces of the casing **100**. Other examples, for reason of design or implementation may be constructed, despite not being explicitly recited herein (for purposes of brevity).

It is appreciated that the increased thermal dissipation provided by the features described herein may have the benefit of reducing (or eliminating) potting, resins, thermosetting plastics, silicone rubber gels, epoxies, and so forth inside the casing **100**. For example, potting might not be applied to any of the elements, such as, the holders **220** or **240**, the first group of inductive elements **320**, or the second group of inductive elements **330**.

Reference is now made to FIG. **5**, which shows a first covering element **510** and second covering element **515** for covering a superstructure. As may be discussed above regarding to FIG. **1**, the casing **100** may be fastened or

6

attached in any other appropriate manner to the superstructure. Fastening, attaching and/or sealing the casing **100** to the superstructure may provide shielding against water, dust, moisture, and so forth (which might otherwise seep into an internal portion of the casing **100** through a seam formed between the casing **100** and the superstructure). FIGS. **5-8** may describe the structural relationship between the casing **100** and the superstructure. The first covering element **510** may be placed over the casing, covering the internal portion **140**. The first covering element **510** may have first hole **520A** and/or a second hole **530A**. The first covering element **510** may also have additional holes, such as holes **550A**. Holes **550A** may be present in the first covering element **510** in order to provide a passage for cabling to pass through. Other holes **540** may be present so that fasteners, such as the kind of fasteners described above, may fasten the first covering element **510** to the casing **100** (or to other parts of the superstructure). The holes **540** may also be used to fasten a circuit board, such as a printed circuit board (not depicted) inside the superstructure. Note that not all such holes which are depicted in FIG. **5** are labeled in the figure in order to reduce visual noise.

The first covering element **510** may be covering the internal portion **140**, and the first hole **520A** may be generally disposed so as to be over the first group of inductive elements **320**. The first covering element **510** may be covering the internal portion **140**, and the second hole **530A** may be generally situated so as to be over the second group of inductive elements **330**. The first hole **520A** and the second hole **530A** may optionally be present in the design of the first covering element **510**. The first covering element **510** may be fully closed (e.g., except for the holes **520A**, **530A** and/or **550A**, such as for passing cabling through the first covering element **510**). The first hole **520A** or the second hole **530A** may be created in the first covering element **510**, and may be relatively large, in order to reduce material used in the first covering element **510**, to reduce the weight of the first covering element **510** and/or to provide easy passage of multiple cables. In some cases, first hole **520A** and/or second hole **530A** may be designed to be small (e.g., similar to the size of holes **550A**, designed to provide a passage for few cables or even a single cable).

The second covering element **515** may be placed in contact the first covering element **510**, such that the first hole **520A** in the first covering element **510** may be generally aligned with the third hole **520B** in the second covering element **515**. The second hole **530A** in the first covering element **510** may be generally aligned with fourth hole **530B** in the second covering element **515**. Holes **550B** in the second covering element **515** may be generally aligned with holes **550A** in the first covering element **510**, and may provide a passage for cabling into the superstructure.

The second covering element **515** may comprise a first wall **560A** and a second wall **560B** which may rise out of the second covering element **515**, and may generally assume the form of a flange beam. Ranges **570A** and **570B** may be used, such as when assembling the superstructure, as support elements.

As may be described below, a second heat dispersing element **580** may be generally disposed on one side of the holes **520B**, **530B**, or **550B**, so as to be generally disposed where the casing **100** is not disposed (e.g., over a second chamber not depicted in FIG. **5**). A space or surface **590** may be provided so that a cooling device (e.g., a fan) which may cool the contents of the superstructure and casing **100** may be added to the superstructure.

The first chamber and the second chamber may be connected to one another with a solid wall disposed between them, such as one of the walls of casing **100**. In some instances, the wall may not be its full height (e.g., it may be only a fraction of its potential height, such as half height, two thirds height, or one quarter height.). In some instances, sections of the wall may be missing (e.g., the wall may be designed to have openings similar to “windows”). In still further aspects, there might not be a wall between the two chambers.

Reference is now made to FIG. 6, which shows different perspectives of elements of a superstructure. A first perspective **610** may be a view of the superstructure from above. A second perspective **620** may be a view of the superstructure from the side. A third perspective **630** may be a view looking at the superstructure from below. In the first perspective **610**, the casing **100** and the first heat dispersing element **130** are both depicted. The first covering element **510** is at the bottom, and may serve as a base for elements displayed in the visible portion of the superstructure in first perspective **610**. A wall **560** (which may comprise one side of the first wall **560A** and the second wall **560B** of FIG. 5) may form at least a portion of one side of the superstructure. Flanges **570** are depicted in the first perspective. A second heat dispersing element **580** may be disposed on one side of the first heat dispersing element **130**. A fan, for example, may be may be affixed to the space or surface **590**, for cooling purposes.

The second heat dispersing element **580** may comprise fins or other appropriate passive or active heat dispersing mechanism. Additionally, the first heat dispersing element **130** or the second heat dispersing element **580** may comprise fins of different lengths, widths, or heights. Gaps between the fins of the first heat dispersing element **130** or the second heat dispersing element **580** may be different sizes. The fins, in the first heat dispersing element **130** or the second heat dispersing element **580**, may be different lengths, heights or widths. The gap between the fins in the first heat dispersing element **130** or the second heat dispersing element **580** may be different lengths, heights or widths. Other, appropriate non-fin type heat dispersing heat dispersing elements, either active or passive, may be used in design of either or both of the first heat dispersing element **130** or the second heat dispersing element **580**. The use of fins in the present disclosure and figures is by way of example only, and not meant in a limiting fashion.

In second perspective **620**, the top and sides of the casing **100** is seen, as well as the first heat dispersing element **130**, which is superficially attached thereto. The first covering element **510** may serve as a base for elements displayed in the visible portion of the superstructure in second perspective **620**. The first wall **560A** and a second wall **560B** are shown, for exemplary purposes, in their position relative to the other elements depicted in the second perspective **620**, as are flanges **570A** and **570B**.

The third perspective **630** may correspond to the bottom of first perspective **610** or second perspective **620**. An arrow (labeled with element number **630**) is depicted in the first perspective **610** and in the second perspective **620**, and may indicate the general position of what is seen in the third perspective **630**. The first covering element **510** is depicted, as are the first hole **520A** and the second hole **530A**. The first group of inductive elements **320** is depicted through the first hole **520**. A portion of the first group of inductive elements **320** and of the second group of inductive elements **330** is depicted through the second hole **530A**. Holes **550A** are also depicted in the third perspective **630**. In some instances, the

first heat dispersing element **130** may extend in more than one direction off of the casing **100**. Such a disposition of the first heat dispersing element **130** may provide dispersion of heat optimized for a particular design of the casing **100** and/or the superstructure. It is appreciated that in other designs or implementations of the casing **100**, the first heat dispersing element **130** may be disposed in a fashion appropriate to a particular design of the casing **100**. In some instances, a space is deliberately left between the first heat dispersing element **130** and the second heat dispersing element **580**. The space may have the benefit of improving heat dispersion. In some instances (e.g., depending on the amount of anticipated heat to be dispersed by the first heat dispersing element **130** and the second heat dispersing element **580**), the first heat dispersing element **130** and the second heat dispersing element **580** may be placed in contact with one another. In some cases, the first heat dispersing element **130** and the second heat dispersing element **580** may be placed in direct contact (e.g., metal-to-metal) with one another, and in some cases, the first heat dispersing element **130** and the second heat dispersing element **580** may be placed in indirect contact (e.g., with a thermal pad inserted between the metallic components) with one another.

Reference is now made to FIG. 7, which shows a structural relationship between a casing (e.g., casing **100** of FIG. 1) and a frame (e.g., frame **700**, which may hold the casing **100** and the superstructure together). The casing **100** may be aligned with the third hole **520B** and the fourth hole **530B**. Dotted lines in FIG. 7 indicate an exemplary physical alignment and placing of the casing **100** into the frame **700** and, therefore, the superstructure. The frame **700** may be designed as part of the superstructure, or the frame **700** may be a separate element used in assembling the casing **100** and the superstructure together.

Reference is now made to FIG. 8, which is a flow diagram **800** of a method for assembling an apparatus included in the casing. It is appreciated that the particular apparatus described is by way of example, and not to be understood in a limiting fashion. Other apparatuses not described herein below may be placed in the casing described herein or its equivalents.

At step **810**, a first heat dispersing element (e.g., a first heat dispersing element **130**) may be disposed superficially on an external surface of a casing (e.g., a casing **100**). The casing may have a hollow space, such as internal portion **140**, formed by an internal surface of the casing. At step **820**, at least one thermal pad, such as insulating padding **210**, may be affixed in the hollow space. At step **830** at least one holder, such as holder **220**, may be attached inside the hollow space. Each holder of the at least one holder may be operative to hold at least one inductive element, such as inductive element **310** or inductive element **340**. At step **840**, at least one inductive element may be placed in the holder. The holder may be disposed such that one inductive element of the at least one inductive element is disposed over at least one of the at least one thermal pad. The casing may be configured to be disposed in a superstructure comprising a second heat dispersing element disposed superficially on an external surface of the superstructure.

Reference is now made to FIG. 9, which is a flow diagram showing an example method for attaching a container (e.g., the container of FIG. 1) to the superstructure. It is appreciated that FIG. 9 may be application specific, and other appropriate methods or individual steps may be performed instead of the steps of FIG. 9 as appropriate. The steps of

FIG. 9 may be performed in a different order (e.g., as may be appropriate to a particular application or implementation).

At step 910, in a container (e.g., a container 100), a first space may be provided for cables from a primary inductive element (e.g., an inductive element 310), and a second space may be provided for cables from a secondary inductive element (e.g., an inductive element 340). At step 920, a covering element (e.g., a covering element 510) may be provided for the container, and the covering element may have at least holes for cables (e.g., holes 550A). At step 930, the covering element may be attached to the container. At step 940, the cables from the primary inductive element and the cables from the secondary inductive element may be passed through the holes in the covering element. At step 950, the covered container may be attached to the superstructure.

Although examples are described above, features and/or steps of those examples may be combined, divided, omitted, rearranged, revised, and/or augmented in any desired manner. For example, step 940 may be performed before step 930. Various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of this description, though not expressly stated herein, and are intended to be within the spirit and scope of the disclosure. Accordingly, the foregoing description is by way of example only, and is not limiting.

The invention claimed is:

1. An apparatus comprising:
 - a casing comprising a first heat dispersing element disposed on an external surface of the casing, wherein the casing comprises a cavity formed by an internal surface of the casing;
 - at least one thermal pad disposed in the cavity, the at least one thermal pad being thermally conductive;
 - at least one inductive element;
 - at least one holder, wherein the holder holds the at least one inductive element; and
 - at least one electrically insulating pad, wherein the at least one thermal pad and the at least one electrically insulating pad are disposed between the at least one inductive element and the internal surface of the casing,
 wherein the at least one holder holds the at least one inductive element over and in thermal contact with the at least one thermal pad, thereby allowing heat generated by the at least one inductive element to flow into the first heat dispersing element.
2. The apparatus of claim 1, further comprising a superstructure, wherein the casing is disposed in the superstructure, wherein the superstructure comprises a second heat dispersing element, wherein the superstructure is in thermal contact with the casing thereby allowing heat generated by the at least one inductive element to flow into the second heat dispersing element.
3. The apparatus of claim 1, wherein the first heat dispersing element comprises fins.
4. The apparatus of claim 1, wherein the at least one holder is made of a thermally insulating material.

5. The apparatus of claim 4, wherein the thermally insulating material comprises an insulating plastic.

6. The apparatus of claim 1, wherein the at least one inductive element comprises an inductor coil.

7. The apparatus of claim 1, wherein the at least one inductive element comprises a choke.

8. The apparatus of claim 1, wherein the at least one inductive element comprises a reactor or a transformer.

9. The apparatus of claim 2, wherein the superstructure comprises: a direct current (DC) to alternating current (AC) inverter, an AC to DC converter, or a DC to DC converter.

10. The apparatus of claim 2, comprising a fan disposed within the superstructure.

11. The apparatus of claim 1, wherein the cavity is free of potting.

12. The apparatus of claim 1, wherein the at least one inductive element comprises a first inductive element and a second inductive element.

13. The apparatus of claim 12, wherein the second inductive element comprises a choke.

14. The apparatus of claim 13, wherein the choke comprises a differential choke.

15. A method comprising:

disposing a first heat dispersing element on an external surface of a casing, the casing having a cavity formed by an internal surface of the casing;

affixing at least one thermally conductive pad in the cavity;

attaching at least one holder in the cavity;

placing at least one inductive element in the at least one holder, the at least one holder being disposed such that the at least one inductive element is disposed over and in thermal contact with the at least one thermally conductive pad, thereby allowing heat generated by the at least one inductive element to flow into the first heat dispersing element; and

placing at least one electrically insulating pad, wherein the at least one thermally conductive pad and the at least one electrically insulating pad are disposed between the at least one inductive element and the internal surface of the casing.

16. The method of claim 15, wherein the method further comprises disposing the casing in a superstructure, wherein the superstructure comprises a second heat dispersing element, wherein the superstructure is in thermal contact with the casing thereby allowing heat generated by the at least one inductive element to flow into the second heat dispersing element.

17. The method of either claim 15, wherein the at least one holder is made of a thermally insulating material.

18. The method of claim 17, wherein the thermally insulating material comprises an insulating plastic.

19. The method of claim 15, wherein the at least one inductive element comprises one or more of: an inductor coil, a choke, a reactor, or a transformer.

20. The method of claim 16, wherein the method further comprises disposing a direct current (DC) to alternating current (AC) inverter, an AC to DC converter, or a DC to DC converter in the superstructure.

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